

English
Medium



EDU TERIA

72nd BPSC

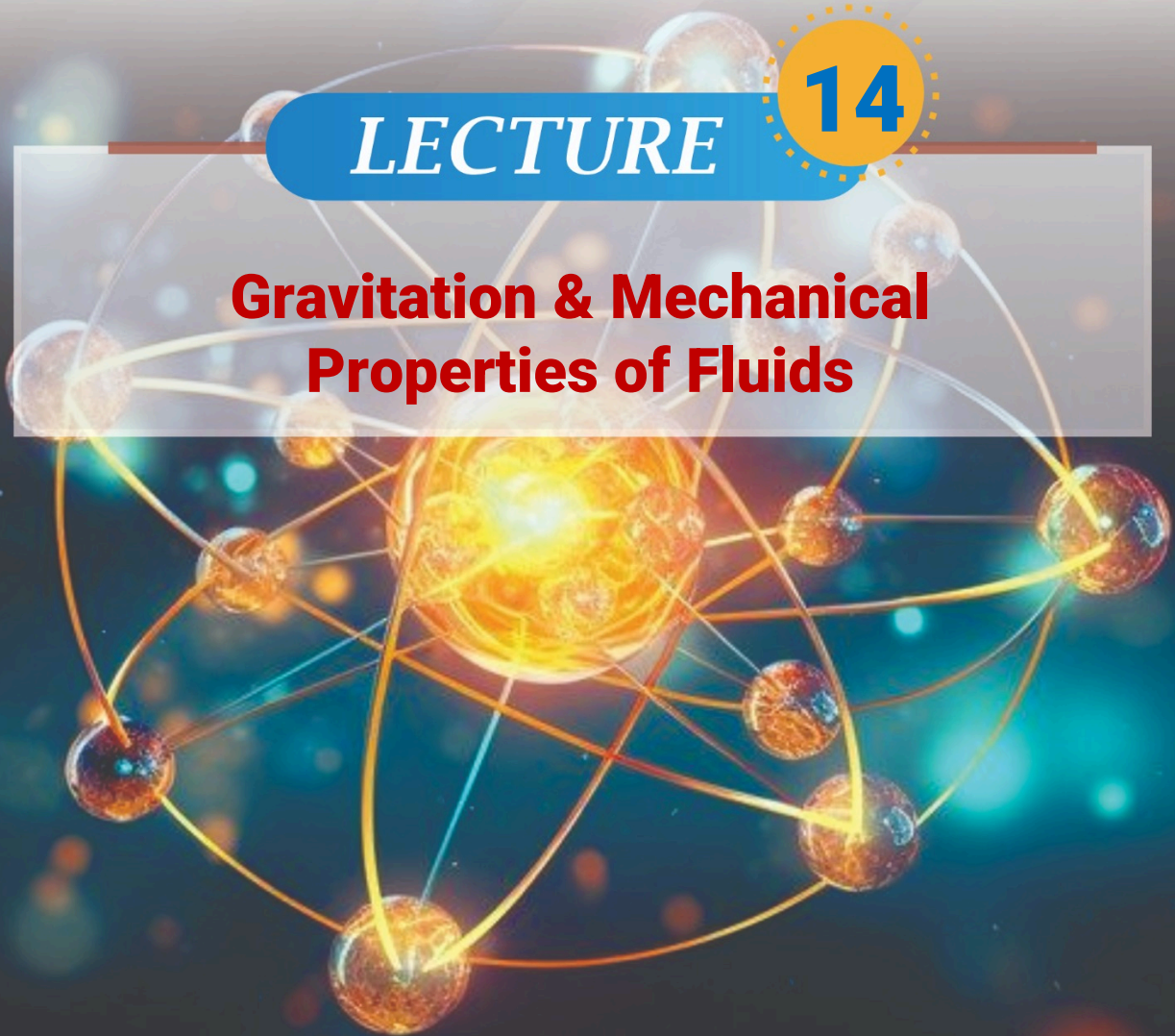
Prelims Online Batch

PHYSICS

LECTURE

14

**Gravitation & Mechanical
Properties of Fluids**



Gravitation & Mechanical Properties of Fluids

Escape Velocity

- It is the minimum velocity required for an object located at the planet's surface so that it just escapes the planet's gravitational field.
- Consider a projectile of mass m , leaving the surface of a planet (or some other astronomical body or system), of radius R and mass M with escape speed V_e .
- When the projectile just escapes to infinity, it has neither kinetic energy nor potential energy.
- Mass (M) and radius (R) of the planet
- Position from where the particle is projected.

Escape speed does not depend on:

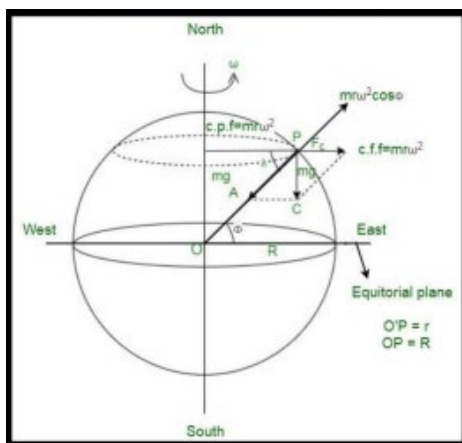
- (i) Mass (m) of the body which is projected
- (ii) Angle of projection.
- If a body is thrown from the Earth's surface with escape speed, it goes out of earth's gravitational field and never returns back to the earth's surface.

Gravitation Force

- Gravitational force is a central as well as conservative force.
- It is the weakest force in nature.
- It is 1036 times smaller than electrostatic force and 10^{-18} times smaller than nuclear force.
- The law of gravitation is applicable for all bodies, irrespective of their size, shape and position.

Factors affecting acceleration due to Gravity

- Shape of Earth Acceleration due to gravity $g \propto 1/R^2$
Earth is elliptical in shape.
- Its diameter at poles is approximately 42 km less than its diameter at equator.
- Therefore, g is minimum at equator and maximum at poles.



Gravitational Field

- The space in the surrounding of anybody in which its gravitational pull
- can be experienced by other bodies is called gravitational fields.

Intensity of Gravitational Field

- The gravitational force acting per unit mass at Earth any point in gravitational field is called intensity of gravitational field at that point.
- It is denoted by E_g or I .
- Intensity of gravitational field at a distance r from a body of mass M is given by
- E_g or $I = GM / r^2$
- It is a vector quantity, and its direction is towards the Centre of gravity of the body.
- Its SI unit is N/m and its dimensional formula is $[LT^{-2}]$.

Escape Velocity

- Escape velocity on earth is the minimum velocity with which a body has to be projected vertically upwards from the earth's surface so that it just crosses the earth's gravitational field and never returns.
- Escape velocity of any object

$$E_g = V_e = \sqrt{2GM/R} = \sqrt{2gR} = \sqrt{8\pi p GR^2/3}$$

- Escape velocity does not depend upon the mass or shape or size of the body as well as the direction of projection of the body.
- Escape velocity at earth is 11.2 km/s.

Some Important Escape Velocities

Heavenly Bodies	Escape Velocities
Moon	2.3 km/s
Mercury	4.28 km/s
Earth	11.2 km/s
Jupiter	60 km/s
Sun	618 km/s
Neutron star	2×10^5 km/s

- Relation between escape velocity and orbital velocity of the satellite

$$E_g = V_e = \sqrt{2}V_0$$



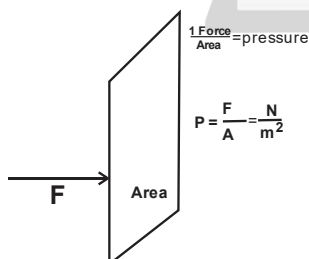
Basis	Mass	Weight
Definition	Amount of matter in a body	Force exerted by gravity on a body
Nature	Scalar quantity	Vector quantity
S.I. Unit	Kilogram (kg)	Newton (N)
Formula	N.A.	$W = mg$
Dependence on gravity	Independent of gravity	Depends on gravity
Value Changes	Same everywhere	Changes with location (Earth, Moon, etc.)
Instrument	Beam balance	Spring balance
Direction	No direction	Always towards center of Earth
Dimension	[M]	[MLT ⁻²]
Zero possibility	Cannot be zero (except no matter)	Can be zero (in space, free fall)

Mechanical Properties of Fluids

- Mechanical properties of fluids (liquids and gases) encompass their behavior at rest (hydrostatics) and in motion (hydrodynamics), defined by pressure, density, viscosity, and surface tension.
- The mechanical properties of fluids (liquids and gases) revolve around their ability to flow, deform under shear stress, and exert pressure.

Fluids

Fluids are substances—specifically liquids and gases—that lack a fixed shape and continuously deform (flow) under applied shear stress.



Pressure (P)

- Force per unit area
- Unit: Pascal (Pa)

Pressure in a Liquid Column:

- Depending on height, density and gravity.
- Formula: $P = h \rho g$

Pascal's Law:

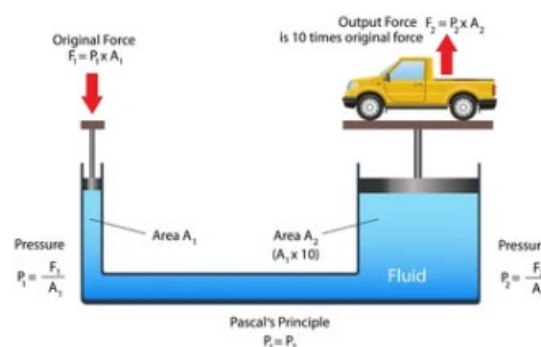
- Pressure applied to a confined fluid is transmitted equally in all directions.
- Used in hydraulic brakes, lifts, jacks.

If the pressure in a liquid is changed at a particular point, the change is transmitted to the entire liquid without being diminished in magnitude. Pascal's law is stated in following ways

- The pressure in a fluid at rest is same at all points if gravity is ignored.
- A liquid exerts equal pressures in all directions.
- If the pressure in an enclosed fluid is changed at a particular point, the change is transmitted to every point of the fluid and to the walls of the container without being diminished in magnitude.

Applications of Pascal's law: hydraulic jacks, lifts, presses, brakes, etc.

- scale velocity does not depend upon the mass or shape or size of the body as well as the direction of projection of the body.



When a body is fully or partially submerged in liquid. It will experience upward force known as Buoyancy force.

Buoyancy force = $\rho g V$

- ρ = density of liquid
- G = Acceleration due to gravity
- V = Volume of body.
- $V = A.h$

Archimedes' Principle:

A body wholly or partially immersed in a fluid experience an upward buoyant force equal to the weight of the fluid displaced by it.

- $F_b = \rho g V$
- Where:
- F_b =buoyant force
- ρ = density of the fluid
- g = acceleration due to gravity

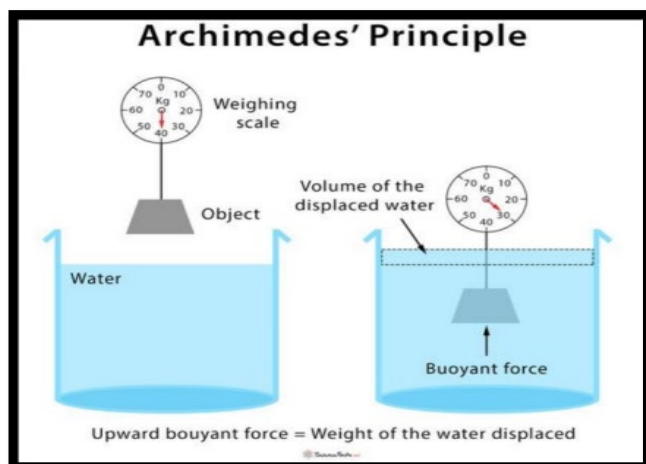


- V = volume of fluid displaced

Buoyant Force: The upward force exerted by fluid is known as the buoyant force or upthrust.

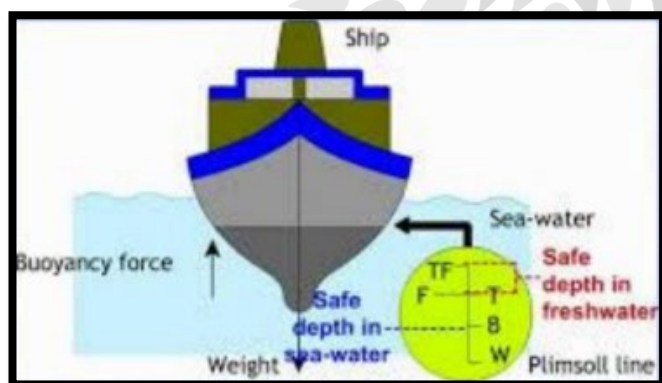
Displaced Fluid: The magnitude of the buoyant force is equal to the weight of the fluid that the object pushes aside.

Density and Floating: An object will float if its average density is less than the density of the fluid it is in, as it will displace a weight of fluid equal to its own weight before becoming fully submerged.



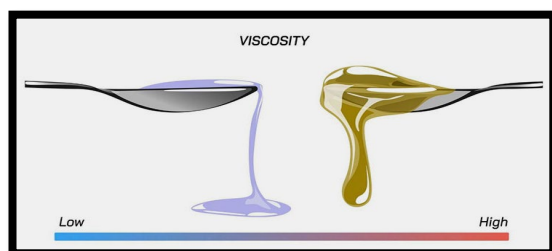
Archimedes' Principal Application:

- Floating ships
- Submarines controlling depth
- Hydrometers (measuring density of liquids)
- Hot air balloons



Viscosity (c)

- Property of liquid and gas.
- Resistance offered by a fluid to flow.
- Viscous force: $F = \eta A dv/dx$
- $F = \frac{\eta A dv}{dx}$, A = Area



Velocity of liquid decreases on increasing temperature liquid.

Temp ↑ = Viscosity ↓ = Flow Increase

Velocity of gas increases on increasing temperature.

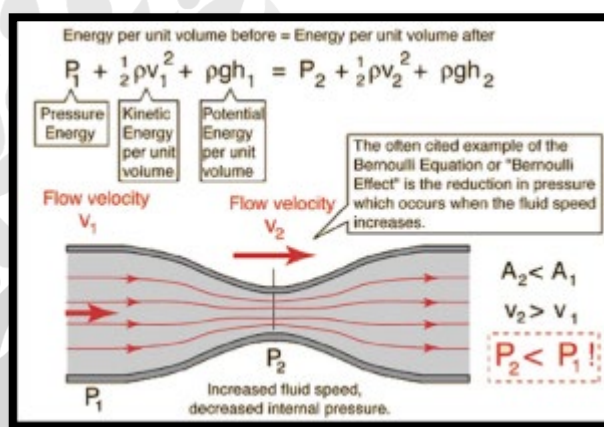
Gas Temp ↑ = Velocity ↑ = Flow resistance ↑

Ideal Fluid

- Steady flow velocity of fluid is constant with respect to time.
- Non-compressible in nature.
- Fluid should be non-viscous.

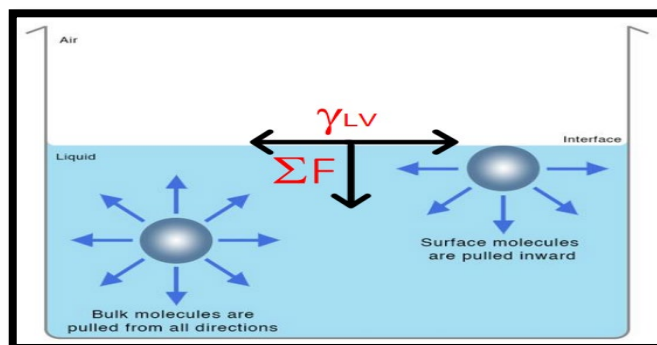
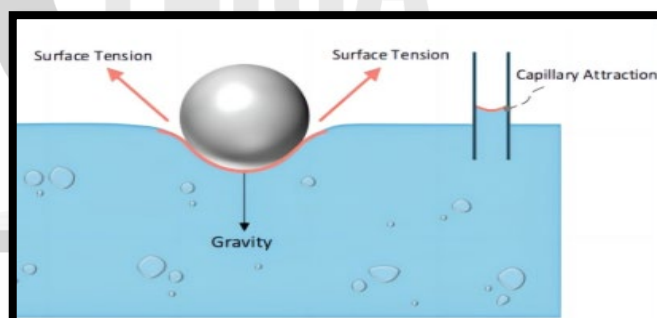
Bernoulli's Equation:

- Based on law of conservation of flowing fluid energy
- For ideal fluid flow (incompressible, non-viscous).
- Formula: $P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$
- Represents conservation of energy.



Surface Tension (T or γ):

- Force acting per unit length along the surface of a liquid.
- Responsible for droplet shape and floating of insects.

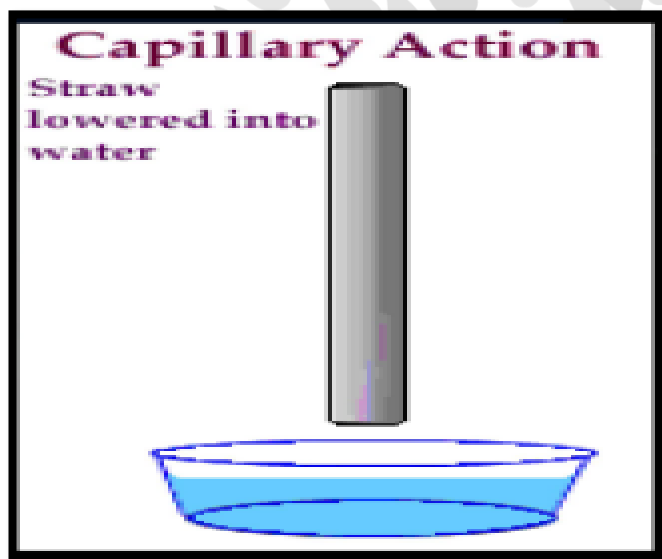
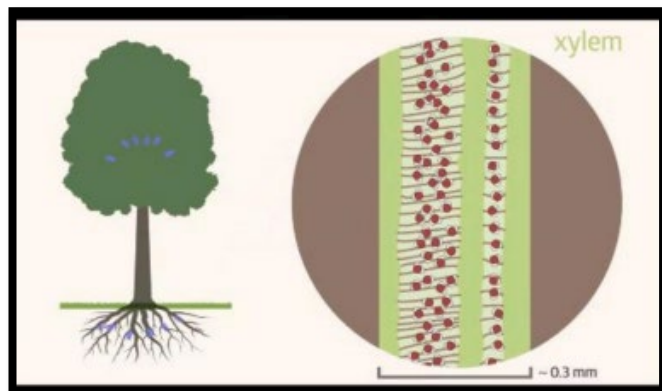


Application of surface vision

- Prolate of water and mercury
- Insects remain floating on liquid surface.

Capillarity:

- Rise or fall of liquid in a narrow tube due to surface tension and angle of contact.
- Example: Water rises in a glass tube.
- Mercury falls in a glass tube.



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